Guidance and Control of Unmanned Autonomous Vehicles

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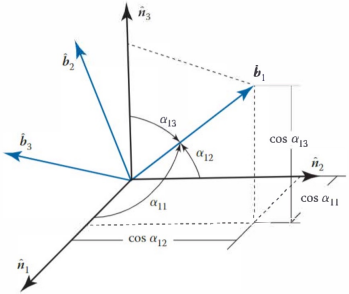
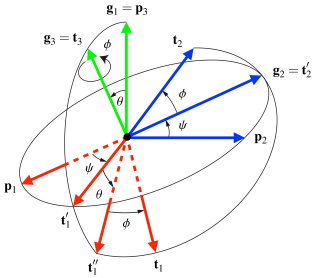
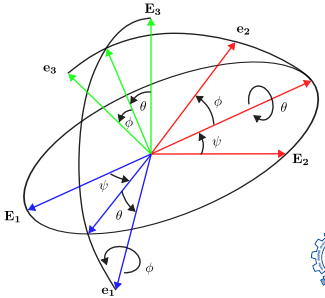
Coordinate Transformation

- **Direction Cosine Matrix**
  - Relation between the orientation of two coordinate frames using 3×3 matrix
- **Euler angles**
  - Method to specify angular orientation of one coordinate frame w.r.t. another
  - A series of three ordered right-handed rotations
  - Correspond to the conventional roll-pitch-yaw angles







Coordinate Transformation

- **Euler angle transformation matrix** $\mathcal{C} = \mathcal{DBA}$

$$\mathcal{C} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

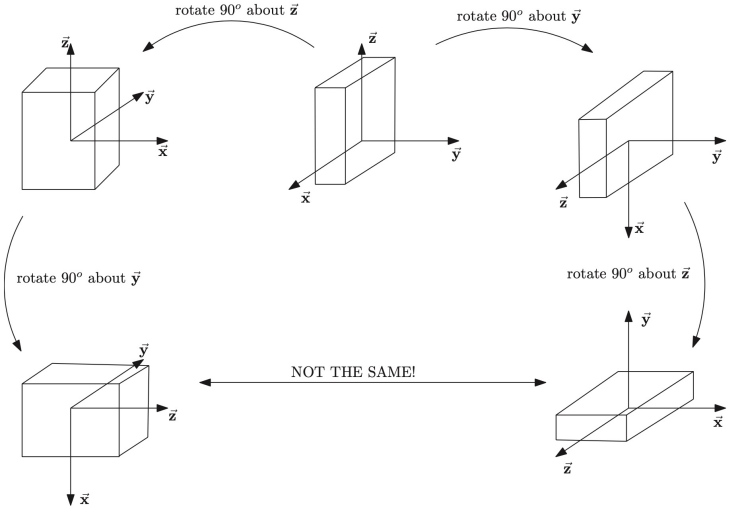
$$= \begin{bmatrix} \cos\theta\cos\psi & \cos\theta\sin\psi & -\sin\theta \\ \cos\psi\sin\theta\sin\phi - \sin\psi\cos\phi & \sin\psi\sin\theta\sin\phi + \cos\psi\cos\phi & \cos\theta\sin\phi \\ \cos\psi\sin\theta\cos\phi + \sin\psi\sin\phi & \sin\psi\sin\theta\cos\phi - \cos\psi\sin\phi & \cos\theta\cos\phi \end{bmatrix}$$

- **Range of Euler angles**:

$$\boxed{-\pi \leq \psi \leq \pi, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}, \quad -\pi \leq \phi \leq \pi}$$

- Is there any issue with range $|\theta| > \pi/2$?

Coordinate Transformation: Order of Rotation



Coordinate Transformation

- Equivalent rotation matrix

$$\mathcal{C} = \begin{bmatrix} \cos\theta\cos\psi & \cos\theta\sin\psi & -\sin\theta \\ \cos\psi\sin\theta\sin\phi - \sin\psi\cos\phi & \sin\psi\sin\theta\sin\phi + \cos\psi\cos\phi & \cos\theta\sin\phi \\ \cos\psi\sin\theta\cos\phi + \sin\psi\sin\phi & \sin\psi\sin\theta\cos\phi - \cos\psi\sin\phi & \cos\theta\cos\phi \end{bmatrix}$$

$$= \begin{bmatrix} \mathcal{C}_{11} & \mathcal{C}_{12} & \mathcal{C}_{13} \\ \mathcal{C}_{21} & \mathcal{C}_{22} & \mathcal{C}_{23} \\ \mathcal{C}_{31} & \mathcal{C}_{32} & \mathcal{C}_{33} \end{bmatrix} = \begin{bmatrix} \boldsymbol{i}^1\cdot\boldsymbol{i}^0 & \boldsymbol{i}^1\cdot\boldsymbol{j}^0 & \boldsymbol{i}^1\cdot\boldsymbol{k}^0 \\ \boldsymbol{j}^1\cdot\boldsymbol{i}^0 & \boldsymbol{j}^1\cdot\boldsymbol{j}^0 & \boldsymbol{j}^1\cdot\boldsymbol{k}^0 \\ \boldsymbol{k}^1\cdot\boldsymbol{i}^0 & \boldsymbol{k}^1\cdot\boldsymbol{j}^0 & \boldsymbol{k}^1\cdot\boldsymbol{k}^0 \end{bmatrix}$$

- How to obtain Euler angles from a given DCM matrix?

$$\theta = \sin^{-1}(-\mathcal{C}_{13}), \quad \phi = \sin^{-1}\left(\frac{\mathcal{C}_{23}}{\sqrt{1-\mathcal{C}_{13}^2}}\right), \quad \psi = \sin^{-1}\left(\frac{\mathcal{C}_{12}}{\sqrt{1-\mathcal{C}_{13}^2}}\right)$$

- Are there some issues with these expressions?

Coordinate Transformation

- Recall the ranges of these Euler angles
- How to determine the quadrant in which these angles lie?
- As pitch angle $\theta \in [-\pi/2, \pi/2]$,

$$\theta \in \begin{cases} [0, \pi/2] & \mathcal{C}_{13} \leq 0 \\ [-\pi/2, 0] & \mathcal{C}_{13} \geq 0 \end{cases}$$

- What about the bank angle ϕ?

- As $\mathcal{C}_{33} = \cos\phi\cos\theta$, and $\cos\theta > 0$, sign of $\mathcal{C}_{33}$ is same as that of $\cos\phi$.
- Also, $\mathcal{C}_{23} = \sin\phi\cos\theta$, and $\cos\theta > 0$, sign of $\mathcal{C}_{23}$ is same as that of $\sin\phi$.

$$\phi \in \begin{cases} \text{First quadrant} & \mathcal{C}_{33} > 0, \quad \mathcal{C}_{23} > 0 \\ \text{Second quadrant} & \mathcal{C}_{33} < 0, \quad \mathcal{C}_{23} > 0 \\ \text{Third quadrant} & \mathcal{C}_{33} < 0, \quad \mathcal{C}_{23} < 0 \\ \text{Fourth quadrant} & \mathcal{C}_{33} > 0, \quad \mathcal{C}_{23} < 0 \end{cases}$$

- **Can we obtain the quadrants of ψ using $\mathcal{C}_{11}$ and $\mathcal{C}_{12}$ in a similar way?** Yes
- What happens to rotation matrix when the angle $\theta = \pm\pi/2$?

Coordinate Transformation

- When angle $\theta = \pi/2$, we have some issues.

$$\mathcal{C} = \begin{bmatrix} \cos\theta\cos\psi & \cos\theta\sin\psi & -\sin\theta \\ \cos\psi\sin\theta\sin\phi - \sin\psi\cos\phi & \sin\psi\sin\theta\sin\phi + \cos\psi\cos\phi & \cos\theta\sin\phi \\ \cos\psi\sin\theta\cos\phi + \sin\psi\sin\phi & \sin\psi\sin\theta\cos\phi - \cos\psi\sin\phi & \cos\theta\cos\phi \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & -1 \\ \cos\psi\sin\phi - \sin\psi\cos\phi & \sin\psi\sin\phi + \cos\psi\cos\phi & 0 \\ \cos\psi\cos\phi + \sin\psi\sin\phi & \sin\psi\cos\phi - \cos\psi\sin\phi & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & -1 \\ \sin(\phi-\psi) & \cos(\phi-\psi) & 0 \\ \cos(\phi-\psi) & -\sin(\phi-\psi) & 0 \end{bmatrix}$$

- **First and third rotations are not distinguishable.**
- What do you expect for $\theta = -\pi/2$? Angle $(\phi + \psi)$ appears in rotation matrix.

Extrinsic vs Intrinsic Rotations (Homework!!)

- **Intrinsic Rotation**: Rotation refers to the last rotated coordinate system
- Equivalent rotation matrix with $3-2-1$ sequence

$$\boxed{\mathcal{C} = \mathcal{R}_{x''}(\phi)\mathcal{R}_{y'}(\theta)\mathcal{R}_z(\psi)}$$

$$\mathcal{C} = \mathcal{R}_{x''}(180^\circ)\mathcal{R}_{y'}(45^\circ)\mathcal{R}_z(90^\circ)$$

- **Extrinsic Rotation**: Rotations refer to a fixed/global coordinate system
- What would be the rotation matrix if these rotations are extrinsic in nature?
- Equivalent rotation matrix with $3-2-1$ sequence

$$\boxed{\mathcal{C} = \mathcal{R}_z(\psi)\mathcal{R}_y(\theta)\mathcal{R}_x(\phi)}$$

$$\mathcal{C} = \mathcal{R}_z(90^\circ)\mathcal{R}_y(45^\circ)\mathcal{R}_x(180^\circ)$$

- How can we obtain the same orientation as through intrinsic rotations?
- Equivalent rotation matrix with $1-2-3$ rotations

$$\boxed{\mathcal{C} = \mathcal{R}_x(\phi)\mathcal{R}_y(\theta)\mathcal{R}_z(\psi)}$$

$$\mathcal{C} = \mathcal{R}_x(180^\circ)\mathcal{R}_y(45^\circ)\mathcal{R}_z(90^\circ)$$

Unmanned Autonomous Vehicle: Simple Rotations

Rotation Matrix

For a given rotation in $\mathbb{R}^3$, there is **one and only one** matrix representing that rotation relative to the initial coordinate frame. Euler angles are, however, not unique.

Simple Rotation

A motion of a rigid body or reference frame B relative to a rigid body or reference frame A is called a simple rotation of B in A if there exists a line L, called an **axis of rotation**, whose orientation relative to both A and B remains unaltered throughout the motion.

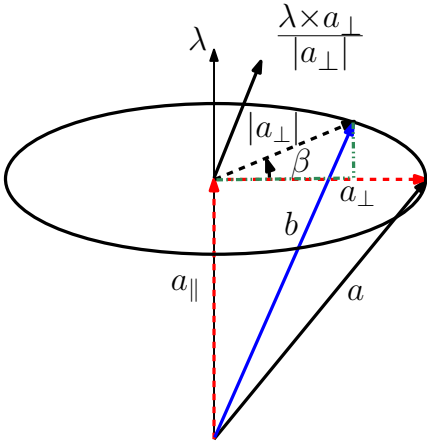
Unmanned Autonomous Vehicles: Coordinate Frame

Euler's Theorem on Rotation

Every change in the relative orientation of two rigid bodies or coordinate frames A and B can be produced by means of a simple rotation of B in A.

- Consider the rotation of a vector in the fixed frame about an arbitrary axis through angle β.
- Let the old and rotated vectors be $\boldsymbol{a}$ and $\boldsymbol{b}$.
- Component of $\boldsymbol{a}$ parallel to axis of rotation $\boldsymbol{\lambda}$

$$\boxed{\boldsymbol{a}_{\parallel} = (\boldsymbol{a} \cdot \boldsymbol{\lambda})\, \boldsymbol{\lambda}}$$



$\boldsymbol{\lambda}$: Unit vector along axis of rotation

Unmanned Autonomous Vehicles: Coordinate Frame

- Component of $\boldsymbol{a} \perp$ to the axis of rotation $\boldsymbol{\lambda}$: $\boxed{\boldsymbol{a}_\perp = \boldsymbol{a} - (\boldsymbol{a} \cdot \boldsymbol{\lambda})\,\boldsymbol{\lambda}}$
- Parallel component, $\boldsymbol{a}_\parallel$, remains unaltered.
- Rotated vector of $\boldsymbol{a}_\perp$ (using $\boldsymbol{\lambda} \perp \boldsymbol{a}_\perp$)

$$\boldsymbol{a}_\perp^{\text{rot}} = \boldsymbol{a}_\perp \cos\beta + |\boldsymbol{a}_\perp| \sin\beta \left(\frac{\boldsymbol{\lambda} \times \boldsymbol{a}_\perp}{|\boldsymbol{a}_\perp|} \right)$$

$$\boldsymbol{a}_\perp^{\text{rot}} = [\boldsymbol{a} - (\boldsymbol{a} \cdot \boldsymbol{\lambda})\,\boldsymbol{\lambda}] \cos\beta + \sin\beta\, \boldsymbol{\lambda} \times [\boldsymbol{a} - (\boldsymbol{a} \cdot \boldsymbol{\lambda})\,\boldsymbol{\lambda}]$$

- Overall rotated vector

$$\boldsymbol{b} = \boldsymbol{a}_\parallel + \boldsymbol{a}_\perp^{\text{rot}} = (\boldsymbol{a} \cdot \boldsymbol{\lambda})\,\boldsymbol{\lambda} + [\boldsymbol{a} - (\boldsymbol{a} \cdot \boldsymbol{\lambda})\,\boldsymbol{\lambda}] \cos\beta + \sin\beta\, \boldsymbol{\lambda} \times [\boldsymbol{a} - (\boldsymbol{a} \cdot \boldsymbol{\lambda})\,\boldsymbol{\lambda}]$$

$$\boxed{\boldsymbol{b} = \boldsymbol{a} \cos\beta + (1 - \cos\beta)\,(\boldsymbol{a} \cdot \boldsymbol{\lambda})\,\boldsymbol{\lambda} + \sin\beta\, \boldsymbol{\lambda} \times \boldsymbol{a}}$$

Unmanned Autonomous Vehicles: Coordinate Frame

- How to obtain the transformation for the case of frame rotation?
- In case of coordinate frame rotation, vector $\boldsymbol{b}$ in new frame in term of $\boldsymbol{a}$ as

$$\boldsymbol{b} = \cos\beta\boldsymbol{a} + (1-\cos\beta)\boldsymbol{\lambda}\boldsymbol{\lambda}^T\boldsymbol{a} - \sin\beta\boldsymbol{\lambda}\times\boldsymbol{a}$$

- Transformation matrix (Rodrigues' Rotation Formula):

$$\boxed{\boldsymbol{b} = \mathcal{C}\boldsymbol{a}} \qquad \boxed{\mathcal{C} = \cos\beta\,\boldsymbol{I} + (1-\cos\beta)\,\boldsymbol{\lambda}\boldsymbol{\lambda}^T - \sin\beta\,\boldsymbol{S}(\boldsymbol{\lambda})}$$

where the skew-symmetric matrix $\boldsymbol{S}(\boldsymbol{\lambda})$ is given by

$$\boldsymbol{S}(\boldsymbol{\lambda}) = \begin{bmatrix} 0 & -\lambda_3 & \lambda_2 \\ \lambda_3 & 0 & -\lambda_1 \\ -\lambda_2 & \lambda_1 & 0 \end{bmatrix}, \quad \boldsymbol{\lambda} = \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix}$$

- For a vector rotation, $\boxed{\mathcal{C} = \cos\beta\,\boldsymbol{I} + (1-\cos\beta)\,\boldsymbol{\lambda}\boldsymbol{\lambda}^T + \sin\beta\,\boldsymbol{S}(\boldsymbol{\lambda})}$

Unmanned Autonomous Vehicles: Coordinate Frame

- Transformation matrix for given axis of rotation ($\boldsymbol{\lambda}$) and angle of rotation (β)

$$
\begin{aligned}
\mathcal{C}_{11} &= (1-\cos\beta)\lambda_1^2 + \cos\beta \\
\mathcal{C}_{22} &= (1-\cos\beta)\lambda_2^2 + \cos\beta \\
\mathcal{C}_{33} &= (1-\cos\beta)\lambda_3^2 + \cos\beta \\
\mathcal{C}_{12} &= (1-\cos\beta)\lambda_1\lambda_2 + \lambda_3\sin\beta \\
\mathcal{C}_{21} &= (1-\cos\beta)\lambda_2\lambda_1 - \lambda_3\sin\beta \\
\mathcal{C}_{23} &= (1-\cos\beta)\lambda_2\lambda_3 + \lambda_1\sin\beta \\
\mathcal{C}_{32} &= (1-\cos\beta)\lambda_3\lambda_2 - \lambda_1\sin\beta \\
\mathcal{C}_{31} &= (1-\cos\beta)\lambda_3\lambda_1 + \lambda_2\sin\beta \\
\mathcal{C}_{13} &= (1-\cos\beta)\lambda_1\lambda_3 - \lambda_2\sin\beta
\end{aligned}
$$

- For $\boldsymbol{\lambda} = (1,0,0)^\top$, $(0,1,0)^\top$, $(0,0,1)^\top$ with angles of rotation, (ϕ,θ,ψ)

$$
\mathcal{C}_{x,\phi} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix}
$$

$$
\mathcal{C}_{y,\theta} = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}
$$

$$
\mathcal{C}_{z,\psi} = \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}
$$

Coordinate Transformation

- Consider the two frames as the a and b frames.
- At time t, the frames a and b are related through the DCM $\mathcal{C}_b^a(t)$.
- At time $t + \Delta t$, the frame b rotates to a new orientation such that the direction cosine matrix is given by $\mathcal{C}_b^a(t + \Delta t)$.
- Rate of change of $\mathcal{C}_b^a(t)$

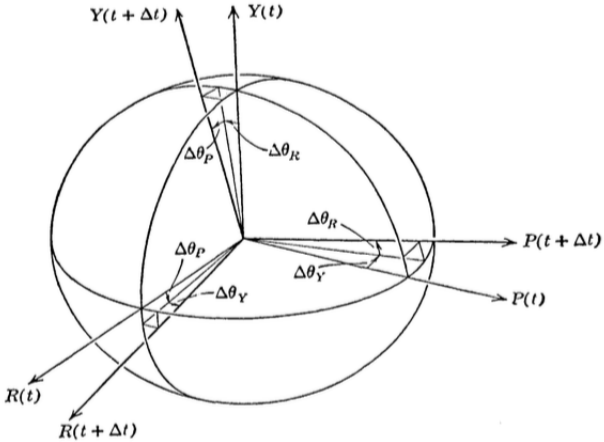
$$\dot{\mathcal{C}}_b^a(t) = \lim_{\Delta t \to 0} \frac{\Delta \mathcal{C}_b^a}{\Delta t} = \lim_{\Delta t \to 0} \frac{\mathcal{C}_b^a(t + \Delta t) - \mathcal{C}_b^a(t)}{\Delta t}$$

- From geometrical considerations,

$$\mathcal{C}_b^a(t + \Delta t) = \mathcal{C}_b^a(t)(\boldsymbol{I} + \boldsymbol{\Delta\theta}^b)$$

where, $\boldsymbol{I} + \boldsymbol{\Delta\theta}^b$ is the small angle DCM relating b frame at time t to the rotated b frame at time $t + \Delta t$.

Coordinate Transformation



- $\boldsymbol{\Delta\theta}^b$ is given by

$$\boldsymbol{\Delta\theta}^b = \begin{bmatrix} 0 & -\Delta\theta_Y & \Delta\theta_P \\ \Delta\theta_Y & 0 & -\Delta\theta_R \\ -\Delta\theta_P & \Delta\theta_R & 0 \end{bmatrix}$$

$$\Delta\theta_k = \sin\Delta\theta_k \ \forall\ k = R, Y, P$$

Coordinate Transformation

- Note that because the rotation angles are small in the limit as $\Delta t \to 0$, small angle approximations are valid and the order of rotation is immaterial.
- Rate of change of $\mathcal{C}_b^i(t)$

$$\dot{\mathcal{C}}_b^a(t) = \mathcal{C}_b^a(t) \lim_{\Delta t \to 0} \frac{\boldsymbol{\Delta\theta}^b}{\Delta t}$$

- In the limit $\Delta t \to 0$, $\boldsymbol{\Delta\theta^b}/\Delta t$ is the skew-symmetric form of angular velocity of the frame b relative to a frame.

$$\dot{\mathcal{C}}_b^a(t) = \mathcal{C}_b^a(t)\boldsymbol{\Omega}_{ab}^b = \mathcal{C}_b^a(t) \begin{bmatrix} 0 & -\omega_Y & \omega_P \\ \omega_Y & 0 & -\omega_R \\ -\omega_P & \omega_R & 0 \end{bmatrix}$$

- **Homework**: Try proving this result using transport theorem or some other ways.

Coordinate Transformation

- DCM differential equation is a linear matrix differential equation, forced by angular velocity vector in its skew-symmetric matrix form.
- Nine scalar, linear, coupled differential equations.
- This equation can be integrated with the initial conditions, which represent the initial orientation of the a-frame w.r.t. the b-frame.
- Differential equation component-wise

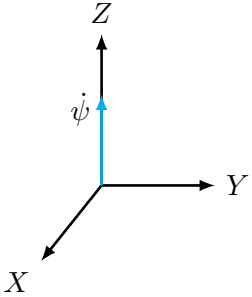
$$\dot{\mathcal{C}}_{i,j} = \mathcal{C}_{i,j+1}\omega_{j+2} - \mathcal{C}_{i,j+2}\omega_{j+1}, \quad i,j = 1,2,3$$

where the second subscript is modulo 3, and

$$\omega_1 = \omega_R \quad \omega_2 = \omega_P, \quad \omega_3 = \omega_Y$$

Coordinate Transformation

- Similar to DCM, Euler angles also vary with time when an input angular velocity vector is applied between two reference frames.
- Angular velocity vector ω, in body-fixed coordinate system, has components p, q, and r in the x, y, and z directions, respectively.

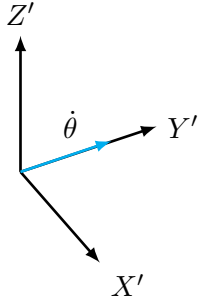


- Consider each derivative of an Euler angle as magnitude of angular velocity vector in coordinate system in which angle is defined.
- $\dot{\psi}$ is magnitude of $\dot{\boldsymbol{\psi}}$ that lies along the Z axis of Earth-fixed coordinate system.

$$\dot{\boldsymbol{\psi}} = \begin{bmatrix} \dot{\psi}_x \\ \dot{\psi}_y \\ \dot{\psi}_z \end{bmatrix} = \mathcal{C} \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} -\dot{\psi}\sin\theta \\ \dot{\psi}\cos\theta\sin\phi \\ \dot{\psi}\cos\theta\cos\phi \end{bmatrix}$$

Coordinate Transformation

- Similarly, the components of $\dot{\boldsymbol{\theta}}$ in $X'Y'Z'$ are given by $(0, \dot{\theta}, 0)^T$.
- In body frame, it can be obtained as



$$\dot{\boldsymbol{\theta}} = \begin{bmatrix} \dot{\theta}_x \\ \dot{\theta}_y \\ \dot{\theta}_z \end{bmatrix} = \mathcal{DB} \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix}$$

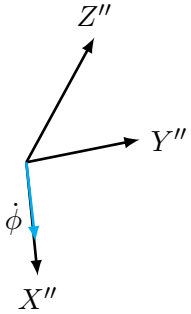
$$\dot{\boldsymbol{\theta}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ \dot{\theta}\cos\phi \\ -\dot{\theta}\sin\phi \end{bmatrix}$$

Coordinate Transformation

- Similarly, the components of $\dot{\boldsymbol{\phi}}$ in $X''Y''Z''$ are given by $(\dot{\phi}, 0, 0)^T$.
- In body frame, it can be obtained as

$$\dot{\boldsymbol{\phi}} = \begin{bmatrix} \dot{\phi}_x \\ \dot{\phi}_y \\ \dot{\phi}_z \end{bmatrix} = \mathcal{D} \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix}$$



- Components of $\boldsymbol{\omega}$ in the body-fixed coordinate system, with suitable unit vectors

$$\boldsymbol{\omega} = \dot{\boldsymbol{\psi}} + \dot{\boldsymbol{\theta}} + \dot{\boldsymbol{\phi}}$$

Coordinate Transformation

- Now, we have body rates given by

$$\begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} \dot{\psi}_x + \dot{\theta}_x + \dot{\phi}_x \\ \dot{\psi}_y + \dot{\theta}_y + \dot{\phi}_y \\ \dot{\psi}_z + \dot{\theta}_z + \dot{\phi}_z \end{bmatrix} = \begin{bmatrix} \dot{\phi} - \dot{\psi}\sin\theta \\ \dot{\psi}\cos\theta\sin\phi + \dot{\theta}\cos\phi \\ \dot{\psi}\cos\theta\cos\phi - \dot{\theta}\sin\phi \end{bmatrix}$$

- Euler angle rates

$$\begin{bmatrix} \dot{\psi} \\ \dot{\theta} \\ \dot{\phi} \end{bmatrix} = \begin{bmatrix} \dfrac{q\sin\phi + r\cos\phi}{\cos\theta} \\ q\cos\phi - r\sin\phi \\ p + \tan\theta(q\sin\phi + r\cos\phi) \end{bmatrix}$$

- What happen when $\theta = \pm 90^\circ$? Gimbal lock problem
- How to avoid such difficulties? Nonsingular representation, e.g., quaternions

Coordinate Transformation

- For $\theta = \pi/2$, $\boxed{p = \dot{\phi} - \dot{\psi},\ q = \dot{\theta}\cos\phi,\ r = -\dot{\theta}\sin\phi}$
- Azimuth and elevation rates

$$\dot{\psi} = \frac{q\sin\phi + r\cos\phi}{\cos\theta} = \frac{\dot{\theta}\cos\phi\sin\phi - \dot{\theta}\sin\phi\cos\phi}{\cos\theta} = \frac{0}{0}$$

$$\dot{\phi} = \ p + \frac{\sin\theta(q\sin\phi + r\cos\phi)}{\cos\theta} = p + \frac{0}{0}$$

Indeterminate forms!!!

- Using L'Hospital rule, and the fact that $\frac{d()}{d\theta} = \frac{d()}{dt}\frac{dt}{d\theta}$, we have

$$\dot{\psi}|_{\theta=\pi/2} = \lim_{\theta\to\pi/2} \frac{\frac{d}{d\theta}\left(q\sin\phi + r\cos\phi\right)}{\frac{d(\cos\theta)}{d\theta}}$$

$$= \lim_{\theta\to\pi/2} \frac{\dot{q}\sin\phi + q\cos\phi\dot{\phi} - r\sin\phi\dot{\phi} + \dot{r}\cos\phi}{-\dot{\theta}\sin\theta} = -\frac{\dot{q}\sin\phi + \dot{r}\cos\phi + \dot{\phi}\dot{\theta}}{\dot{\theta}}$$

Coordinate Transformation

- Also, for $\theta = \pi/2$, $\boxed{p = \dot{\phi} - \dot{\psi}}$

$$\dot{\phi}|_{\theta=\pi/2} = p + \dot{\psi}|_{\theta=\pi/2} = \; p - \frac{\dot{q}\sin\phi + \dot{r}\cos\phi + \dot{\phi}\dot{\theta}}{\dot{\theta}}$$

- On solving this equation,

$$\dot{\phi}|_{\theta=\pi/2} = \frac{p}{2} - \frac{\dot{q}\sin\phi + \dot{r}\cos\phi}{2\dot{\theta}}$$

- Also, $\dot{\theta} = q\cos\phi - r\sin\phi$.
- For $\theta \approx \pi/2$, use these limiting values, else use usual update equations.
- What will happen to these relations between Euler angle rates and body rates if the sequence of rotations are changed?
- Are these relation invariant w.r.t. the sequence of rotations?

Coordinate Transformation

- How to obtain these relations for different orders of rotations?
- Find the relation if Euler angle sequence of $X - Y' - Z''$, with Euler angles (ϕ, θ, ψ) is considered.
- Relation between Euler angle rates and body rates

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} \dot{\phi}_x + \dot{\theta}_x + \dot{\psi}_x \\ \dot{\phi}_y + \dot{\theta}_y + \dot{\psi}_y \\ \dot{\phi}_z + \dot{\theta}_z + \dot{\psi}_z \end{bmatrix} = \mathcal{ABD} \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix} + \mathcal{AB} \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} + \mathcal{A} \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix}$$

$$= \begin{bmatrix} \dot{\phi} \cos\psi \cos\theta + \dot{\theta} \sin\psi \\ -\dot{\phi} \cos\theta \sin\psi + \dot{\theta} \cos\psi \\ \dot{\phi} \sin\theta + \dot{\psi} \end{bmatrix}$$

- **Be careful with matrices here for their correct order of multiplications**
- **Homework**: **Find out the relation for sequence** $Y - X' - Z$"

Rotation Matrix: Properties

A 3×3 matrix is a rotation operator in $\mathbb{R}^2$ **if and only if** it is an orthogonal matrix and has determinant $+1$.

- Rotation about axis $\boldsymbol{v}$ by angle θ
- Scalar product of two vectors is invariant under rotation.

$$(\boldsymbol{A}\boldsymbol{v}_1)^\top(\boldsymbol{A}\boldsymbol{v}_2) = \boldsymbol{v}_1^\top\boldsymbol{v}_2$$

$$\boldsymbol{v}_1^\top\boldsymbol{A}^\top\boldsymbol{A}\boldsymbol{v}_2 = \boldsymbol{v}_1^\top\boldsymbol{v}_2$$

$$\boldsymbol{v}_1^\top(\boldsymbol{A}^\top\boldsymbol{A} - \boldsymbol{I})\boldsymbol{v}_2 = 0 \implies \boldsymbol{A}^\top\boldsymbol{A} = \boldsymbol{I}$$

- As $|\boldsymbol{A}| = |\boldsymbol{A}^\top|$, we have

$$|\boldsymbol{A}|^2 = |\boldsymbol{A}||\boldsymbol{A}^\top| = |\boldsymbol{A}\boldsymbol{A}^\top| = |\boldsymbol{I}| = 1$$

- Thus, $|\boldsymbol{A}| = \pm1$.
- As rotation preserve the right handedness of coordinate frame $|\boldsymbol{A}| \neq -1$, and $|\boldsymbol{A}| = 1$.

$$\boldsymbol{B} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}. \; |\boldsymbol{B}| = -1.$$

Rotation Matrix: Properties

- Suppose $\boldsymbol{A}$ is orthogonal and $|\boldsymbol{A}| = 1$.
- Axis of any rotation is fixed under rotation.
- We need to show $\boldsymbol{A}\boldsymbol{v}_0 = \boldsymbol{v}_0$.
- $\boldsymbol{A}$ must have a characteristic value or eigenvalue of 1, $|\boldsymbol{A} - \boldsymbol{I}| = 0$.

$$\begin{aligned} |\boldsymbol{A} - \boldsymbol{I}| &= |\boldsymbol{A}^\top||\boldsymbol{A} - \boldsymbol{I}| = |\boldsymbol{A}^\top\boldsymbol{A} - \boldsymbol{A}^\top| \\ &= |\boldsymbol{I} - \boldsymbol{A}^\top| = |(\boldsymbol{I} - \boldsymbol{A})^\top| = |\boldsymbol{I} - \boldsymbol{A}| \\ &= -|\boldsymbol{A} - \boldsymbol{I}| \end{aligned}$$

- Thus, $|\boldsymbol{A} - \boldsymbol{I}| = 0$.
- Existence of $\boldsymbol{v}_0$ as axis of rotation.

- Need to show that any vector in the plane perpendicular to $\boldsymbol{v}_0$ is mapped by $\boldsymbol{A}$ into another vector in that plane.
- $\boxed{\boldsymbol{A}\boldsymbol{v}_0 = \boldsymbol{v}_0 \implies \boldsymbol{A}^\top\boldsymbol{v}_0 = \boldsymbol{v}_0}$

$$\boldsymbol{A}^\top\boldsymbol{v}_0 = \boldsymbol{A}^\top\boldsymbol{A}\boldsymbol{v}_0 = \boldsymbol{I}\boldsymbol{v}_0 = \boldsymbol{v}_0$$

- With $\boldsymbol{v}$ in plane $\perp \boldsymbol{v}_0$, $\boldsymbol{v}^\top\boldsymbol{v}_0 = 0$

$$(\boldsymbol{A}\boldsymbol{v})^\top\boldsymbol{v}_0 = \boldsymbol{v}^\top\boldsymbol{A}^\top\boldsymbol{A}\boldsymbol{v}_0 = \boldsymbol{v}^\top\boldsymbol{v}_0 = 0.$$

- With orthogonal $\boldsymbol{A}$, scalar product in invariant under this transformation.
- $\boldsymbol{A}$ must be rotation operator.

Rotation Matrix: Properties

- How to find the axis of rotation for given rotation matrix $\boldsymbol{A}$?
- Every rotation matrix has a real eigenvalue of $+1$, $(\boldsymbol{A}-\boldsymbol{I})\boldsymbol{v}=\boldsymbol{0}$.

$$
\begin{aligned}
(a_{11}-1)v_1 + a_{12}v_2 + a_{13}v_3 &= 0 \\
a_{21}v_1 + (a_{22}-1)v_2 + a_{23}v_3 &= 0 \\
a_{31}v_1 + a_{32}v_2 + (a_{33}-1)v_3 &= 0
\end{aligned}
$$

- With $v_3 = 1$,

$$
\begin{aligned}
a_{21}v_1 + (a_{22}-1)v_2 &= -\,a_{23} \\
a_{31}v_1 + a_{32}v_2 &= -\,(a_{33}-1)
\end{aligned}
$$

- Possible solution

$$
\boldsymbol{v} = \begin{bmatrix} a_{12}a_{23} - (a_{22}-1)a_{13} \\ a_{21}a_{13} - (a_{11}-1)a_{23} \\ (a_{11}-1)(a_{22}-1) - a_{12}a_{21} \end{bmatrix}
$$

- Amount of rotation:

$$
\phi = \cos^{-1}\left[\frac{\text{Trace}(\boldsymbol{A})-1}{2}\right]
$$

- Is there any simpler way to obtain this axis of rotation?

Rotation Matrix: Properties

- Axis $\boldsymbol{v}$ is eigenvector with $\lambda = 1$.
- Nontrivial solution of

$$(\boldsymbol{A} - \lambda \boldsymbol{I})\boldsymbol{v} = \boldsymbol{0} \text{ iff } |\boldsymbol{A} - \lambda \boldsymbol{I}| = 0$$

- Solution for $(\boldsymbol{A} - \boldsymbol{I})\boldsymbol{v} = \boldsymbol{0}$
- As $\boxed{\boldsymbol{v} = \boldsymbol{A}\boldsymbol{v} = \boldsymbol{A}^\top \boldsymbol{v}}$

$$\left(\boldsymbol{A} - \boldsymbol{A}^\top\right)\boldsymbol{v} = \boldsymbol{0} \implies \boldsymbol{Q}\boldsymbol{v} = \boldsymbol{0}$$

$$\begin{bmatrix} 0 & q_3 & -q_2 \\ -q_3 & 0 & q_1 \\ q_2 & -q_1 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

- With $v_3 = 1$, $q_3 v_2 - q_2 = 0$, $\quad q_3 v_1 - q_1 = 0$

$$\boldsymbol{v}^\top = \begin{bmatrix} \dfrac{q_1}{q_3} & \dfrac{q_2}{q_3} & 1 \end{bmatrix} \equiv \begin{bmatrix} q_1 & q_2 & q_3 \end{bmatrix}$$

If a rotation is defined by a matrix

$$\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

then its fixed axis of rotation is given by

$$\boldsymbol{v} = \boldsymbol{i}(a_{23} - a_{32}) + \boldsymbol{j}(a_{31} - a_{13}) + \boldsymbol{k}(a_{12} - a_{21}).$$

Text/References

References

- George M. Siouris, Aerospace Avionics Systems: A Modern Synthesis, Academic Press, Inc. 1993.
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Thank you for your attention !!!